



Optimization-Based Image Processing

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Image optimization problem

$$\operatorname{argmin}_{\mathbf{x}} \sum_{i=1}^I f_i(\mathbf{K}_i \mathbf{x}) \quad \text{with} \quad \mathbf{K} = \begin{bmatrix} \mathbf{K}_1 \\ \vdots \\ \mathbf{K}_I \end{bmatrix}$$

Denoising (WNNM)

$$\min_{\mathbf{X}} \|\mathbf{Y} - \mathbf{X}\|_F^2 + \|\mathbf{X}\|_{\mathbf{w},*}$$

TVL1 deconvolution

$$\mathbf{E}(\mathbf{I}) = \|\mathbf{I} \otimes k - \mathbf{B}\| + \lambda \|\nabla \mathbf{I}\|_{tv}$$

Multi-image SR

$$\hat{\mathbf{x}} = \arg \min \left[\sum_{k=1}^p \|\mathbf{y} - \mathbf{W}_k \hat{\mathbf{x}}\|^2 + \alpha \sum_{c \in S} \varphi_c(\mathbf{x}) \right]$$



Optimization problem solver

- Closed-form solution for quadratic formulation
- Problem-specific iterative algorithm, e.g.
 - Richard-Lucy deconvolution for Poisson formulation
 - Belief propagation for Markov Random Field
- Generic iterative algorithm, e.g.
 - **Half quadratic splitting**
 - **Primal-dual algorithm**
 - Alternating direction method of multiplier (ADMM)



Outline

- **Half quadratic splitting (HQS)**
 - Image optimization with denoiser prior
 - Fast TVL1 deconvolution
- Primal-dual algorithm
 - ProxImaL language
- Application examples

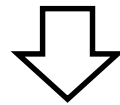
Additive HQS

[D. Geman et al, IEEE TIP, 1995]



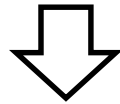
HQS for image optimization problem

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \underbrace{\frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2}_{\text{data term}} + \underbrace{\lambda \Phi(\mathbf{x})}_{\text{regularization term}}$$



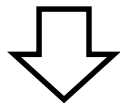
Split problems by an auxiliary $\mathbf{z}$

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2 + \lambda \Phi(\mathbf{z}) \quad s.t. \quad \mathbf{z} = \mathbf{x}$$



New cost function

$$\mathcal{L}_{\mu}(\mathbf{x}, \mathbf{z}) = \frac{1}{2} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2 + \lambda \Phi(\mathbf{z}) + \frac{\mu}{2} \|\mathbf{z} - \mathbf{x}\|^2$$



Iterative solution between two problems

$$\begin{cases} \mathbf{x}_{k+1} = \arg \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2 + \mu \|\mathbf{x} - \mathbf{z}_k\|^2 \\ \mathbf{z}_{k+1} = \arg \min_{\mathbf{z}} \frac{\mu}{2} \|\mathbf{z} - \mathbf{x}_{k+1}\|^2 + \lambda \Phi(\mathbf{z}) \end{cases}$$

Data iteration:
Given $\mathbf{z}$ to update $\mathbf{x}$

Regularization iteration:
Given $\mathbf{x}$ to update $\mathbf{z}$

Ref: K. Zhang, et. al., "Learning deep CNN denoiser prior for image restoration," CVPR, 2017.



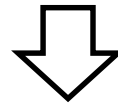
Denoiser prior for image optimization

$$\mathbf{x}_{k+1} = (\mathbf{H}^T \mathbf{H} + \mu \mathbf{I})^{-1} (\mathbf{H}^T \mathbf{y} + \mu \mathbf{z}_k)$$

Data iteration has analytic solution

$$\mathbf{z}_{k+1} = \arg \min_{\mathbf{z}} \frac{1}{2(\sqrt{\lambda/\mu})^2} \|\mathbf{x}_{k+1} - \mathbf{z}\|^2 + \Phi(\mathbf{z})$$

Regularization iteration depends on the regularization prior



Can be treated by any denoising technique,
e.g. BM3D, NLM, CNN, ...

$$\mathbf{z}_{k+1} = \text{Denoiser}(\mathbf{x}_{k+1}, \sqrt{\lambda/\mu})$$

Image optimization with different measurement matrices $\mathbf{H}$, e.g. deblurring, super-resolution, and demosaic, can be solved in the same framework with denoiser prior.



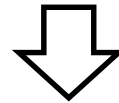
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- Half quadratic splitting (HQS)
 - Image optimization with denoiser prior
 - **Fast TVL1 deconvolution**
- Primal-dual algorithm
 - ProxImaL language
- Application examples



HQS for TVL1 deconvolution

$$E(I) = \|I \otimes k - B\| + \lambda \|\nabla I\|_{tv}$$



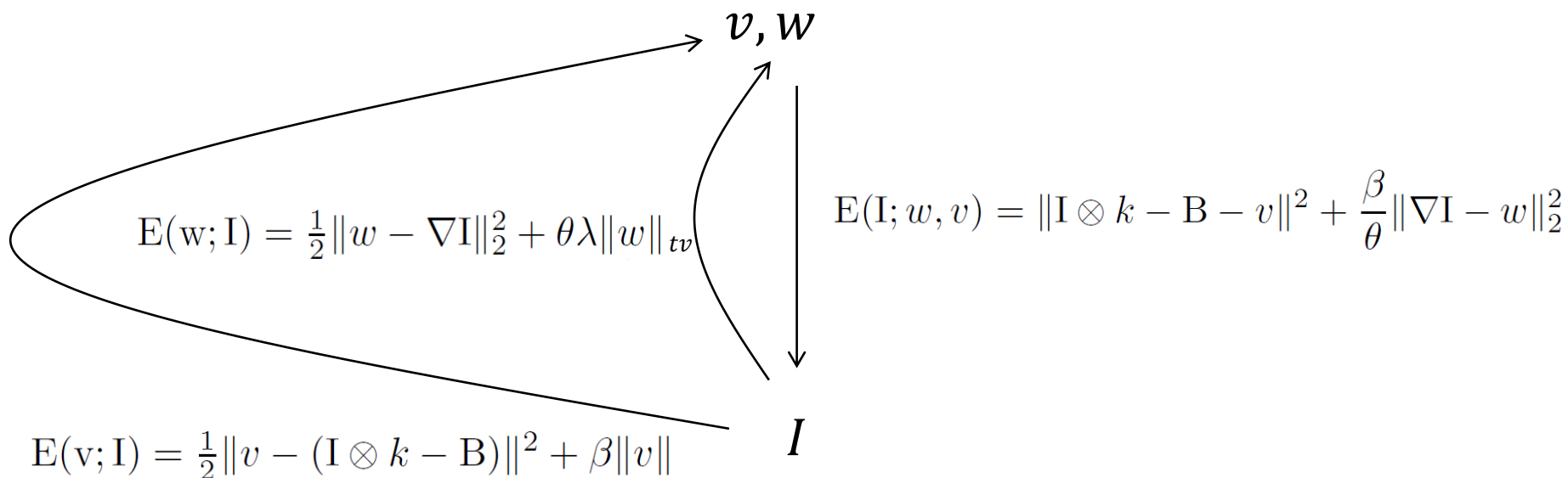
Split problems by auxiliary v and w

$$E(I, w, v) = \frac{1}{2\beta} \|I \otimes k - B - v\|^2 + \frac{1}{2\theta} \|\nabla I - w\|_2^2 + \|v\| + \lambda \|w\|_{tv}$$

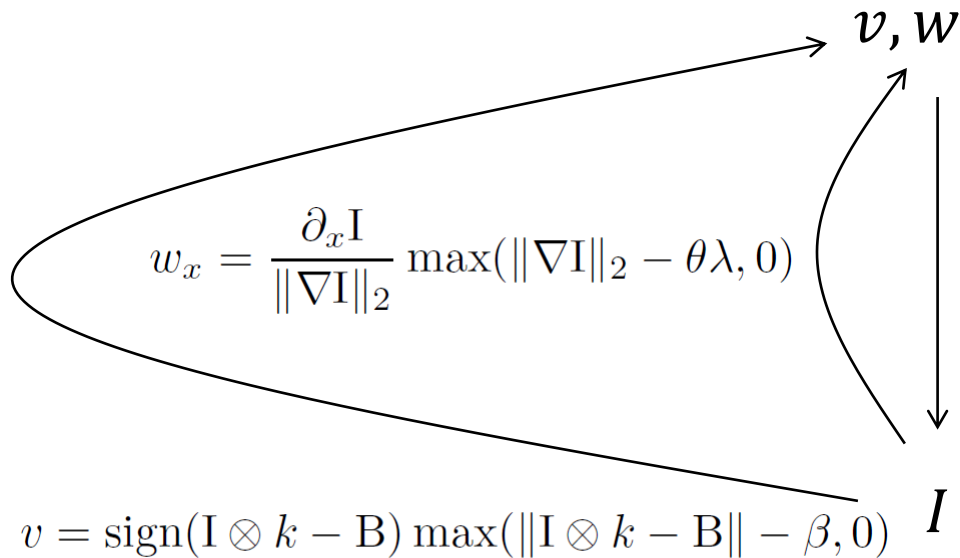
When $\beta \rightarrow 0$ and $\theta \rightarrow 0$, the solution of the below equation approaches to that of the original one.



Iterative algorithm



Fast algorithm



$$\mathcal{F}(I) = \frac{\overline{\mathcal{F}(k)} \mathcal{F}(B + v) + \beta / \theta (\overline{\mathcal{F}(\partial_x)} \mathcal{F}(w_x) + \overline{\mathcal{F}(\partial_y)} \mathcal{F}(w_y))}{\overline{\mathcal{F}(k)} \mathcal{F}(k) + \beta / \theta (\overline{\mathcal{F}(\partial_x)} \mathcal{F}(\partial_x) + \overline{\mathcal{F}(\partial_y)} \mathcal{F}(\partial_y))}$$

Algorithm 3. Robust Deconvolution

INPUT: Blurred image B and the estimated kernel k^s
 Edge taping in Matlab
 $I \leftarrow B, \beta \leftarrow \beta_0$.
repeat
 Solve for v using Eq. (18)
 $\theta \leftarrow \theta_0$
 repeat
 Solve for w using Eq. (17)
 Solve for I in the frequency domain using Eq. (15)
 $\theta \leftarrow \theta/2$
 until $\theta < \theta_{\min}$
 $\beta \leftarrow \beta/2$
until $\beta < \beta_{\min}$
OUTPUT: Deblurred image I

Deconvolution result



(a) Noisy input



(b) [15]

6 minutes



(c) [17]

1.5 seconds
(fast hyper-laplacian prior)



(d) Ours

6 seconds



Outline

- Half quadratic splitting (HQS)
 - Image optimization with denoiser prior
 - Fast TVL1 deconvolution Fast but unable to guarantee convergence
- **Primal-dual algorithm**
 - **ProxImaL language** Guarantee convergence for convex functions
- Application examples

Ref:

F. Heide et al, "ProxImaL: Efficient image optimization using proximal algorithms," SIGGRAPH, 2016.

F. Heide et al, "High-quality computational imaging through simple lenses," ACM TOG, 2013.

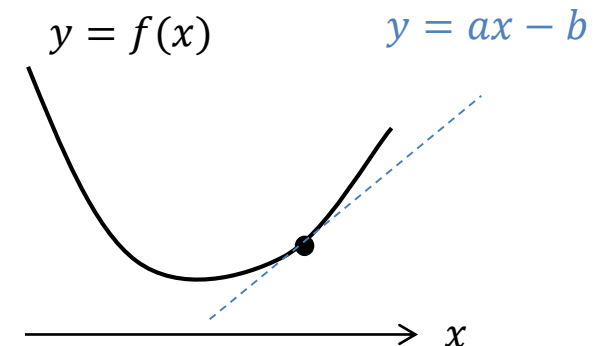


Convex conjugate

- Real-valued convex function $f(x)$:
 - Bijective/invertible mapping between variable x and slope a
 - $x = h(a), h = f'^{-1}$
 - Convex conjugate function $b = a\hat{x} - f(\hat{x}) := f^*(a), \hat{x} = f^{-1}(a)$
 - also convex
- Extension for multi-variate $f(\mathbf{x})$:
 - Bijective/invertible mapping between variable $\mathbf{x}$ and normal $\mathbf{a}$
 - Convex conjugate function $b = \langle \mathbf{a}, \hat{\mathbf{x}} \rangle - f(\hat{\mathbf{x}}) := f^*(\mathbf{a})$ (also convex)
- Property: $f^{**} = f$

Example:

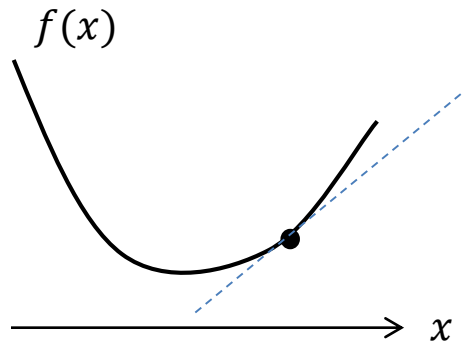
$$f(x) = e^x;$$
$$f^*(x^*) = x^* \log(x^*) - x^*.$$





Definition with inequality

Convex conjugate $f^*(x^*)$ of convex function $f(x)$:



For each x , we have a tangent function $h(x) = \langle a, x \rangle - b$ such that $f(x) \geq h(x) \Rightarrow b \geq \langle a, x \rangle - f(x)$

$$x, f(x) \xleftrightarrow{1\text{-to-1}} x^* = a, f^*(x^*) = b$$

Inequality definition

$$f^*(x^*) = \max_x \langle x^*, x \rangle - f(x)$$

⇓ Conjugate symmetry $f^{**} = f$

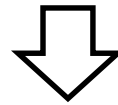
$$f(x) = \max_{x^*} \langle x, x^* \rangle - f^*(x^*)$$

We can find $f(x)$ with given f^* via maximization.



Primal-dual problem splitting

$$\min_{\mathbf{x}} \underbrace{G(\mathbf{x})}_{\text{data term}} + \underbrace{F(\mathbf{K}\mathbf{x})}_{\text{regularization term}}$$



Generic saddle-point problem
via convex conjugate with an auxiliary $\mathbf{y}$

$f \rightarrow F, x \rightarrow Kx, x^* \rightarrow y$
from previous slide

$$\min_{\mathbf{x}} \max_{\mathbf{y}} G(\mathbf{x}) + (\langle \mathbf{K}\mathbf{x}, \mathbf{y} \rangle - F^*(\mathbf{y}))$$

$F^*(\mathbf{y})$ is convex conjugate of
convex function $F(\mathbf{K}\mathbf{x})$

$$F(\mathbf{K}\mathbf{x}) = \max_{\mathbf{y}} \langle \mathbf{K}\mathbf{x}, \mathbf{y} \rangle - F^*(\mathbf{y})$$



First-order primal-dual algorithm (Pock-Chambolle algorithm)

$$\begin{cases} \mathbf{y}_{n+1} &= \text{prox}_{\sigma F^*}(\mathbf{y}_n + \sigma \mathbf{K} \bar{\mathbf{x}}_n) & \text{regularization iteration} \\ \mathbf{x}_{n+1} &= \text{prox}_{\tau G}(\mathbf{x}_n + \tau \mathbf{K}^* \mathbf{y}_{n+1}) & \text{data iteration} \\ \bar{\mathbf{x}}_{n+1} &= \mathbf{x}_{n+1} + \theta (\mathbf{x}_{n+1} - \mathbf{x}_n) \end{cases}$$



Proximal operator

- Pock-Chambolle algorithm defines

$$\begin{aligned}\mathbf{prox}_{\tau G}(\tilde{\mathbf{x}}) &= \underset{\mathbf{x}}{\operatorname{argmin}} \frac{1}{2\tau} \|\mathbf{x} - \tilde{\mathbf{x}}\|_2^2 + \mathbf{G}(\mathbf{x}) \\ &:= (\mathbb{I} + \tau \partial \mathbf{G})^{-1}(\tilde{\mathbf{x}})\end{aligned}$$

Given previous estimation $\tilde{x}$, refine a new x based on constraint $G(x)$

Example of generalization

For a demosaic problem:

$$\mathbf{x}_{\text{opt}} = \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{MD}\mathbf{x} - \mathbf{b}\|_2^2 + r(\mathbf{x})$$

$$r(\mathbf{x}) = \mu \|\nabla \mathbf{x}\|_1 + (1 - \mu) \|\nabla \mathbf{x}\|_2^2 + I_{[0,\infty)}(\mathbf{x})$$

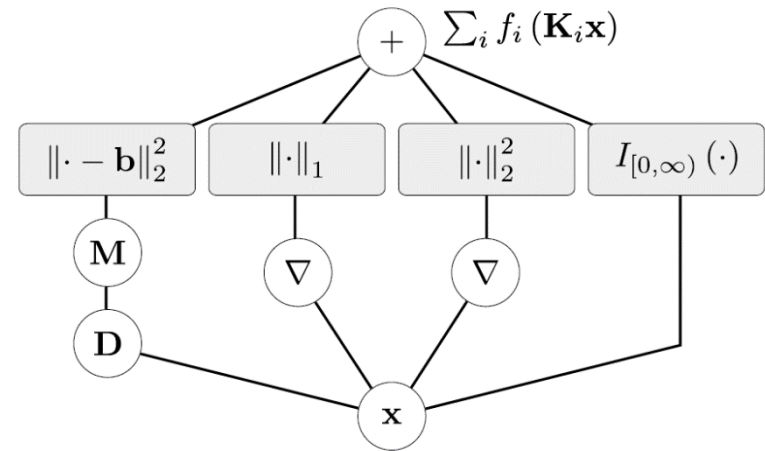
$$f_1(\mathbf{v}) = \|\mathbf{v} - \mathbf{b}\|_2^2, \quad \mathbf{K}_1 = \mathbf{MD}$$

$$f_2(\mathbf{v}) = \mu \|\mathbf{v}\|_1, \quad \mathbf{K}_2 = \nabla$$

$$f_3(\mathbf{v}) = (1 - \mu) \|\mathbf{v}\|_2^2, \quad \mathbf{K}_3 = \nabla$$

$$f_4(\mathbf{v}) = I_{[0,\infty)}(\mathbf{v}), \quad \mathbf{K}_4 = \mathbf{I}$$

model:



Algorithm 1 Pock-Chambolle to solve Problem (2)

```

1: Initialization:  $\sigma\tau\|\mathbf{K}\|_2^2 < 1$ ,  $\theta \in [0, 1]$ ,  $(\mathbf{x}^0, \mathbf{z}^0)$ ,  $\bar{\mathbf{x}}^0 = \mathbf{x}^0$ .
2: for  $k = 1$  to  $V$  do
3:    $\mathbf{z}_j^{k+1/2} = \mathbf{z}_j^k + \sigma\mathbf{K}_j\bar{\mathbf{x}}^k \quad \forall j \in \Psi$ 
4:    $\mathbf{z}_j^{k+1} = \mathbf{z}_j^{k+1/2} - \sigma\mathbf{prox}_{f_j/\sigma}(\mathbf{z}_j^{k+1/2}/\sigma) \quad \forall j \in \Psi$ 
5:   if  $\Omega = \{f_i\}$  then
6:      $\mathbf{x}^{k+1} = \mathbf{prox}_{\tau f_i}(\mathbf{x}^k - \tau\mathbf{K}^T\mathbf{z}^{k+1})$ 
7:   else
8:      $\mathbf{x}^{k+1} = \mathbf{x}^k - \tau\mathbf{K}^T\mathbf{z}^{k+1}$ 
9:   end if
10:   $\bar{\mathbf{x}}^{k+1} = \mathbf{x}^{k+1} + \theta(\mathbf{x}^{k+1} - \mathbf{x}^k)$ 
11: end for

```

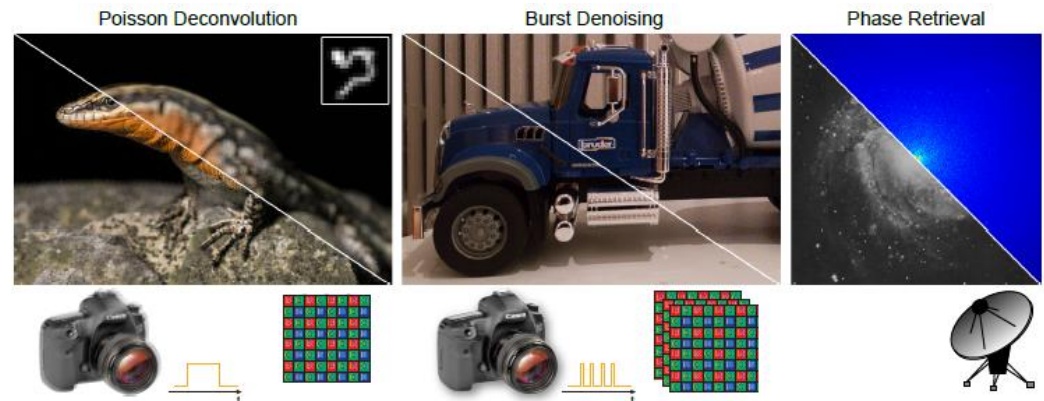
ProxImaL language

- Allow users to describe problems using a simple domain-specific language (in Python) and also compile them
 - Compiler: problem rewriting, splitting, solver implementation
 - Limit: proximal functions

```
x = Variable(300, 300, 3)
data_term = sum_squares( subsample( conv(x, psf) ) - input )
grad_term = mu * norm1( grad(x) ) +
            (1-mu) * sum_squares( grad(x) )
objective = data_term + grad_term + nonneg(x)
p = Problem( objective )
```

p.solve()

Solvers include HQS, PC, ADMM.

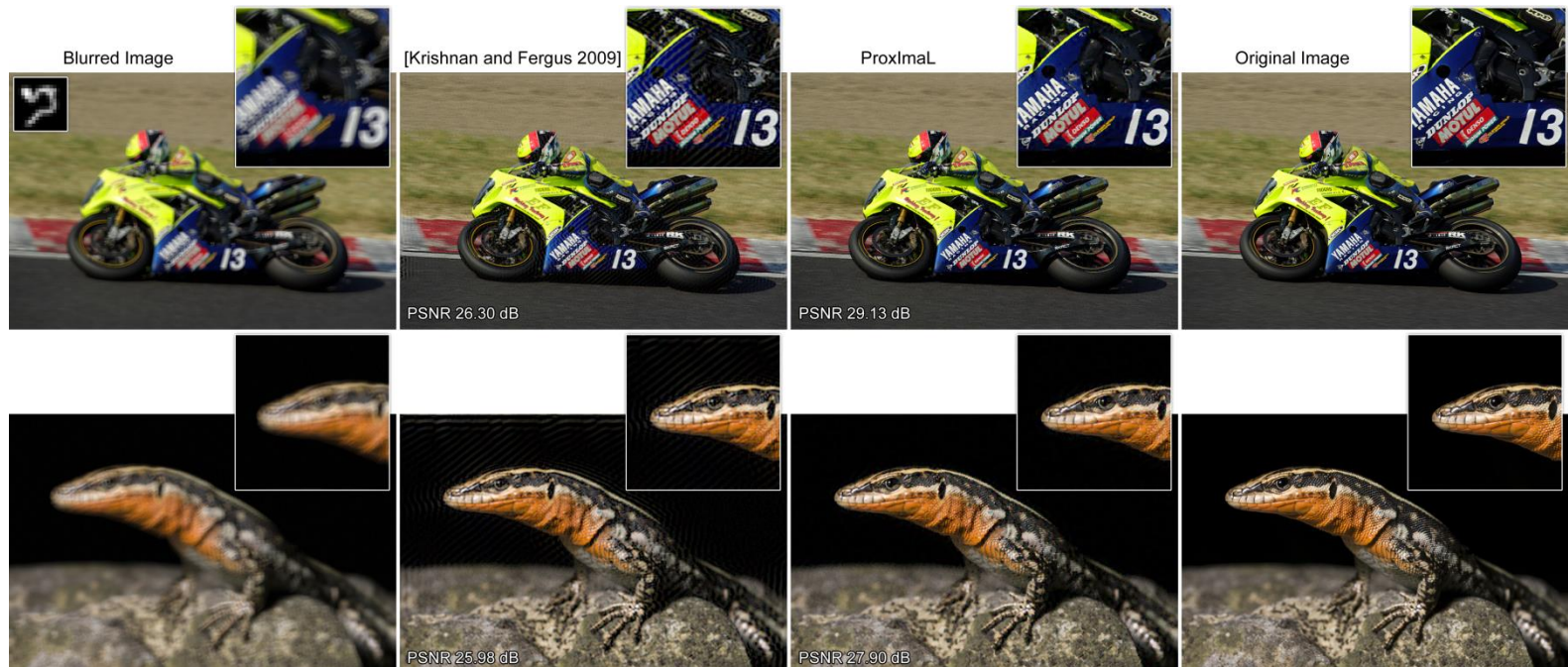
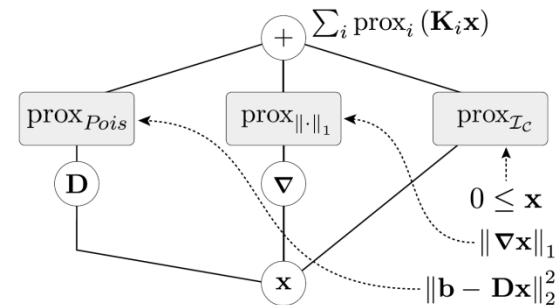


Example: Poisson deconvolution

```

x = Variable(300, 300, 3)
data_term = poisson_norm( conv(x, psf) - input )
grad_sparsity = norm1( grad(x) )
objective = data_term + grad_sparsity + nonneg(x)
p = Problem( objective )

```

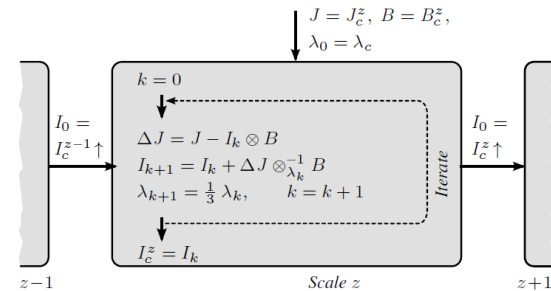




Outline

- Half quadratic splitting (HQS) Fast
 - Image optimization with denoiser prior
 - Fast TVL1 deconvolution
- Primal-dual algorithm Convergence guaranteed
 - ProxImaL language High-level language to define and solve problems
- Application examples
 - High-quality simple-lenses imaging
 - FlexISP (flexible image signal processing)
 - Lensless imaging

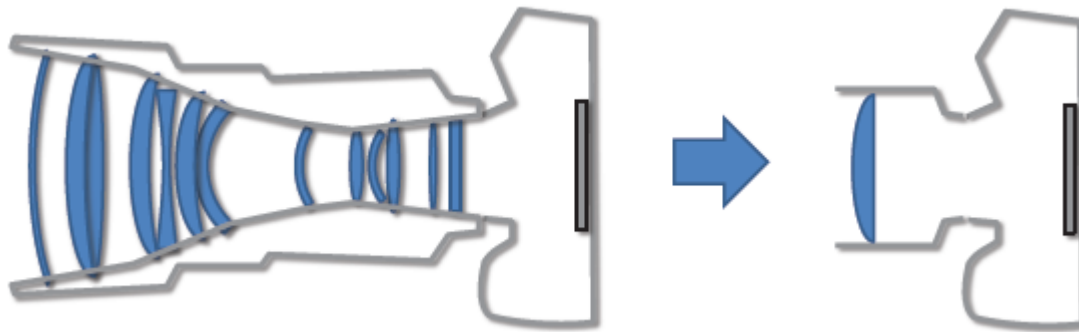
Application: Sharp imaging through simple lenses



Ref: F. Heide, et. al., "High-quality computational imaging through simple lenses," ACM TOG, 2013.

Sharp imaging through simple lenses

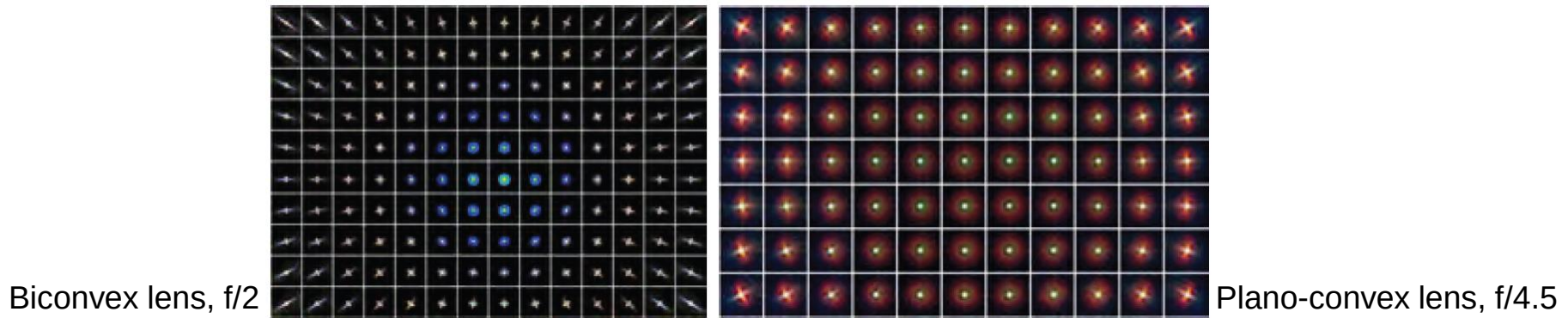
- Idea
 - A (off-line) robust estimation of per-channel spatially-varying point spread function
 - A **cross-channel prior** for color images
 - A deconvolution method using a convex solver which guarantees convergence (**primal-dual** algorithm)



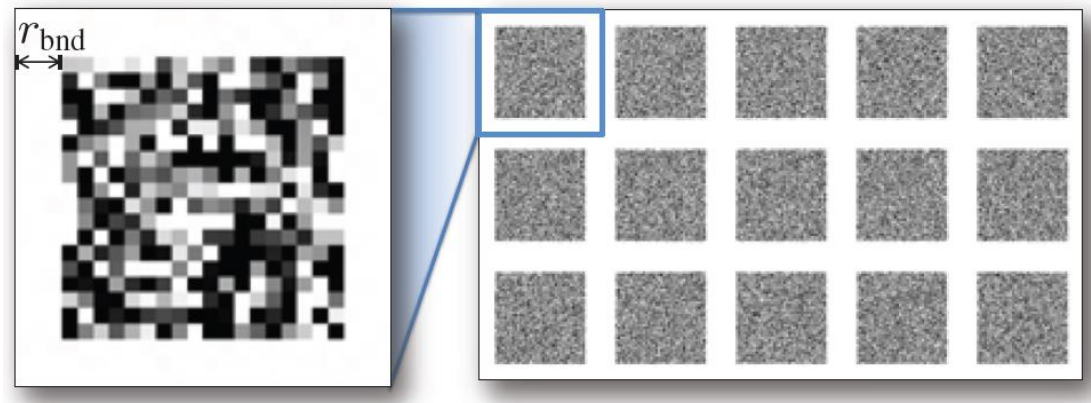
Ref: F. Heide, et. al., “High-quality computational imaging through simple lenses,” ACM TOG, 2013.

Per-channel spatially-varying PSF

- Patch-wise estimation
 - Measure non-linearity of optics (not defocus blur)



- Calibration pattern



Deconvolution

- Problem

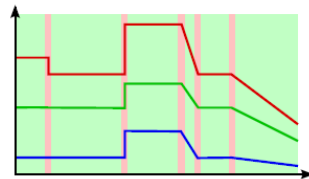
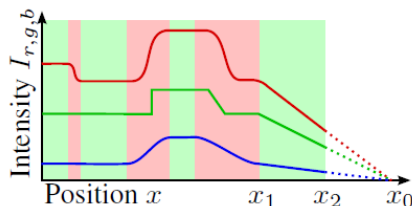
$$J = B \otimes I + N$$

$$\mathbf{j} = \mathbf{B}\mathbf{i} + \mathbf{n},$$

- Optimization formulation

$$(\mathbf{i}_{1\dots 3})_{\text{opt}} = \underset{\mathbf{i}_{1\dots 3}}{\text{argmin}} \sum_{c=1}^3 \|\mathbf{B}_c \mathbf{i}_c - \mathbf{j}_c\|_2^2 + \lambda_c \sum_{a=1}^5 \|\mathbf{H}_a \mathbf{i}_c\|_1$$

$$+ \sum_{l \neq c} \beta_{cl} \sum_{a=1}^2 \|\mathbf{H}_a \mathbf{i}_c \cdot \mathbf{i}_l - \mathbf{H}_a \mathbf{i}_l \cdot \mathbf{i}_c\|_1,$$



Cross-channel term
(keep sharpness across channels)

$$\begin{aligned} \nabla \mathbf{i}_k / \mathbf{i}_k &\approx \nabla \mathbf{i}_l / \mathbf{i}_l \\ \Leftrightarrow \nabla \mathbf{i}_k \cdot \mathbf{i}_l &\approx \nabla \mathbf{i}_l \cdot \mathbf{i}_k \end{aligned}$$



Primal-dual algorithm

$$\begin{aligned}
 (\mathbf{i}_c)_{\text{opt}} &= \underset{\mathbf{x}}{\operatorname{argmin}} \left\| \begin{bmatrix} \lambda_c \mathbf{H}_1 \\ \vdots \\ \lambda_c \mathbf{H}_5 \\ \beta_{cl} (\mathbf{D}_{i_l} \mathbf{H}_1 - \mathbf{D}_{H_1 i_l}) \\ \vdots \\ \beta_{cl} (\mathbf{D}_{i_l} \mathbf{H}_2 - \mathbf{D}_{H_2 i_l}) \\ \vdots \end{bmatrix} \mathbf{x} \right\|_1 + \|\mathbf{B}_c \mathbf{x} - \mathbf{j}_c\|_2^2 \\
 &= \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{S} \mathbf{x}\|_1 + \|\mathbf{B}_c \mathbf{x} - \mathbf{j}_c\|_2^2
 \end{aligned} \tag{10}$$

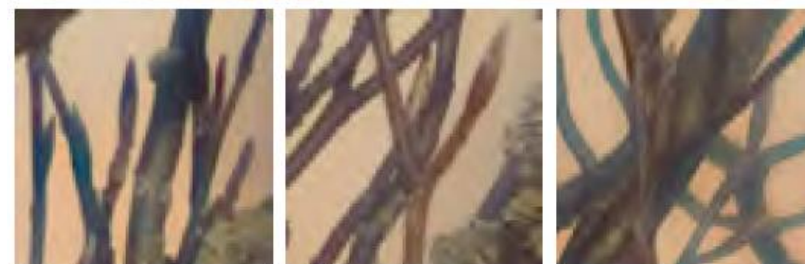
$$\mathbf{K}(\mathbf{x}) = \mathbf{S} \mathbf{x}$$

$$\mathbf{F}(\mathbf{y}) = \|\mathbf{y}\|_1$$

$$\mathbf{G}(\mathbf{x}) = \|\mathbf{B}_c \mathbf{x} - \mathbf{j}_c\|_2^2$$

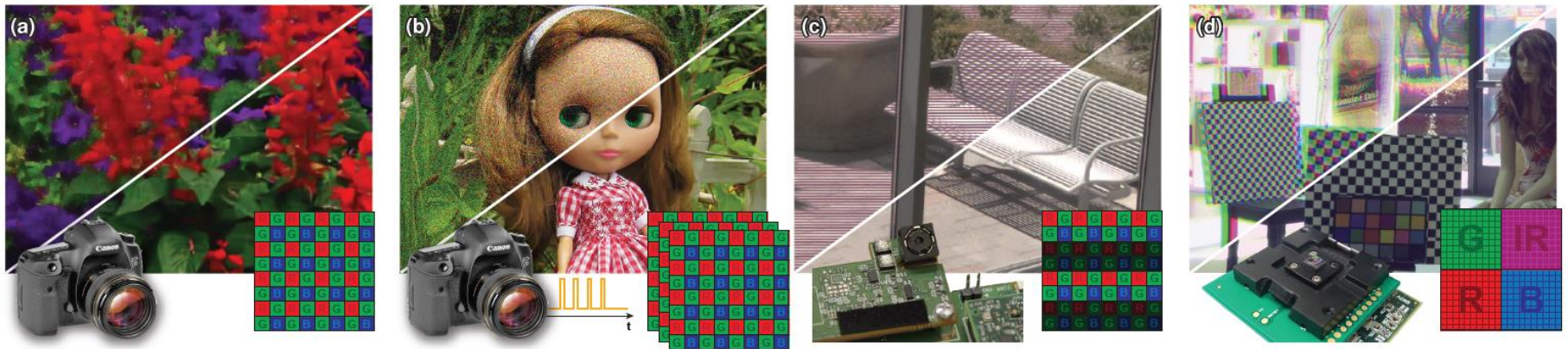
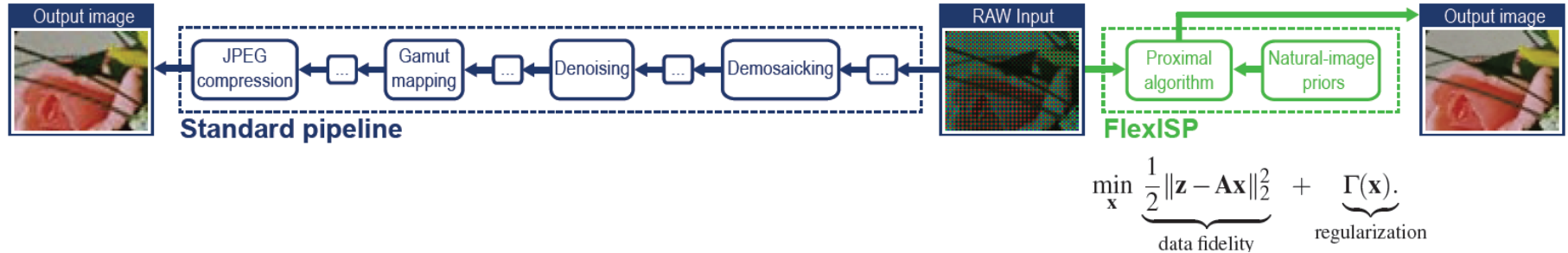
$$\begin{aligned}
 \mathbf{y} &= \operatorname{prox}_{\sigma \mathbf{F}^*}(\tilde{\mathbf{y}}) \Leftrightarrow y_i = \frac{\tilde{y}_i}{\max(1, |\tilde{y}_i|)} \\
 \mathbf{x} &= \operatorname{prox}_{\tau \mathbf{G}}(\tilde{\mathbf{x}}) \\
 \Leftrightarrow 0 &= \frac{1}{\tau} (\mathbf{x} - \tilde{\mathbf{x}}) + 2 (\mathbf{B}_c^T \mathbf{B}_c \mathbf{x} - \mathbf{B}_c^T \mathbf{j}) \\
 \Leftrightarrow \mathbf{x} &= \mathcal{F}^{-1} \left(\frac{\tau 2 \mathcal{F}(\mathbf{B}_c)^* \mathcal{F}(\mathbf{j}_c) + \mathcal{F}(\tilde{\mathbf{x}})}{\tau 2 |\mathcal{F}(\mathbf{B}_c)|^2 + 1} \right),
 \end{aligned}$$

Result



Application: FlexISP

- Flexible and integrated image processing framework
 - Use conjugate gradient + denoising prior for primal-dual algorithm
 - Claim better performance for individual specific techniques



Ref: F. Heide, et. al., "FlexISP: A flexible camera image processing framework," SIGGRAPH Asia, 2014.

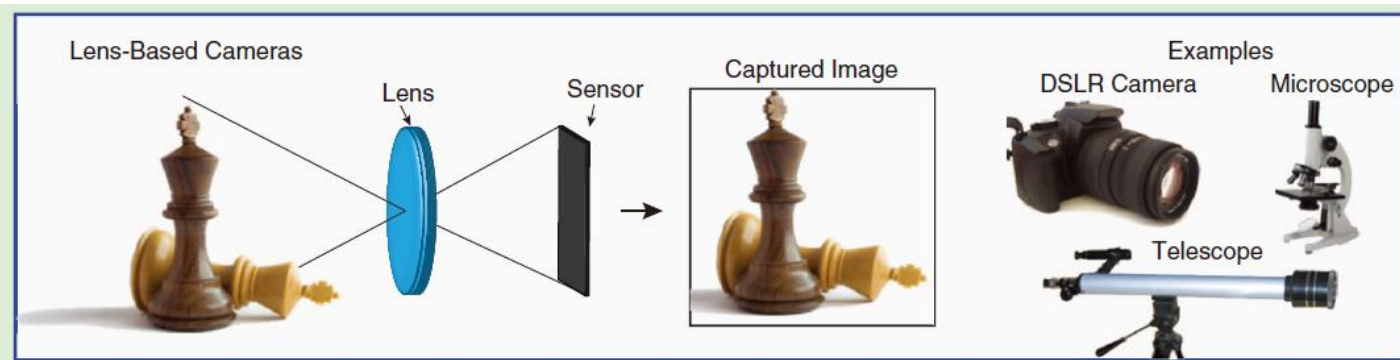


Outline

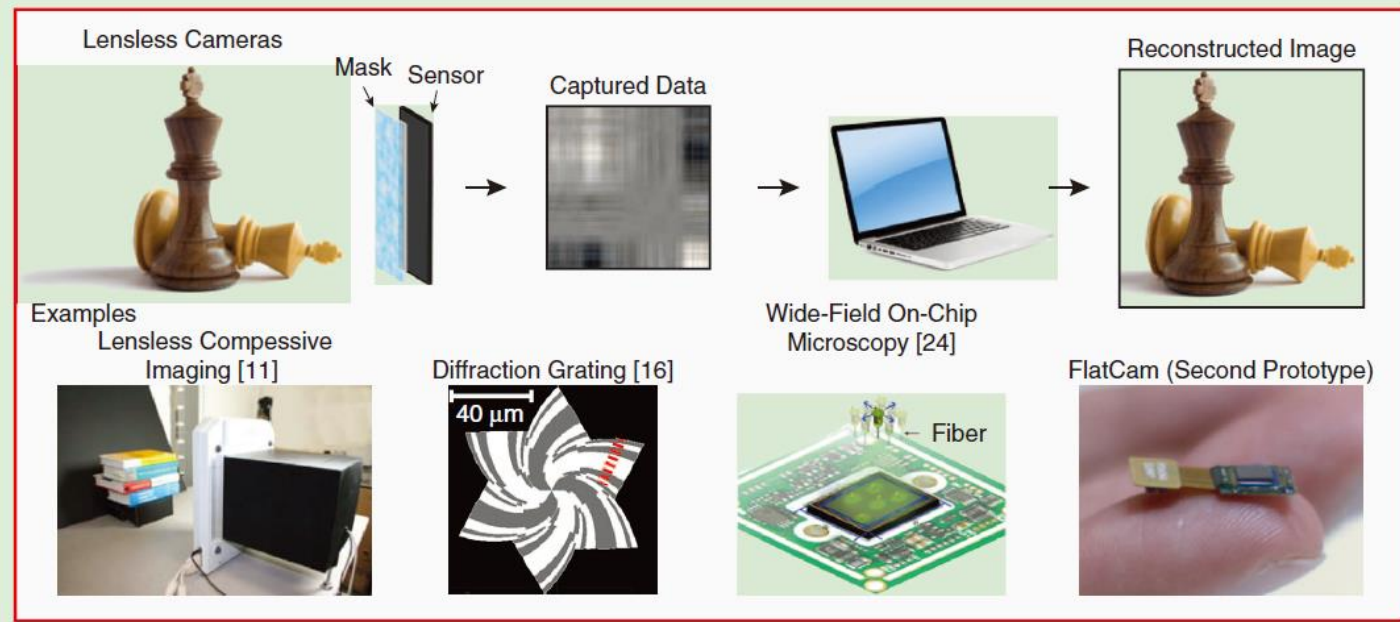
- Half quadratic splitting (HQS)
- Primal-dual algorithm
- Application examples
 - High-quality simple-lenses imaging
 - FlexISP (flexible image signal processing)
 - **Lensless imaging**
 - FlatCam
 - Diffractive optics

Ref: V. Boominathan, et. al., "Lensless imaging," *IEEE Signal Processing Mag.*, Sept 2016.

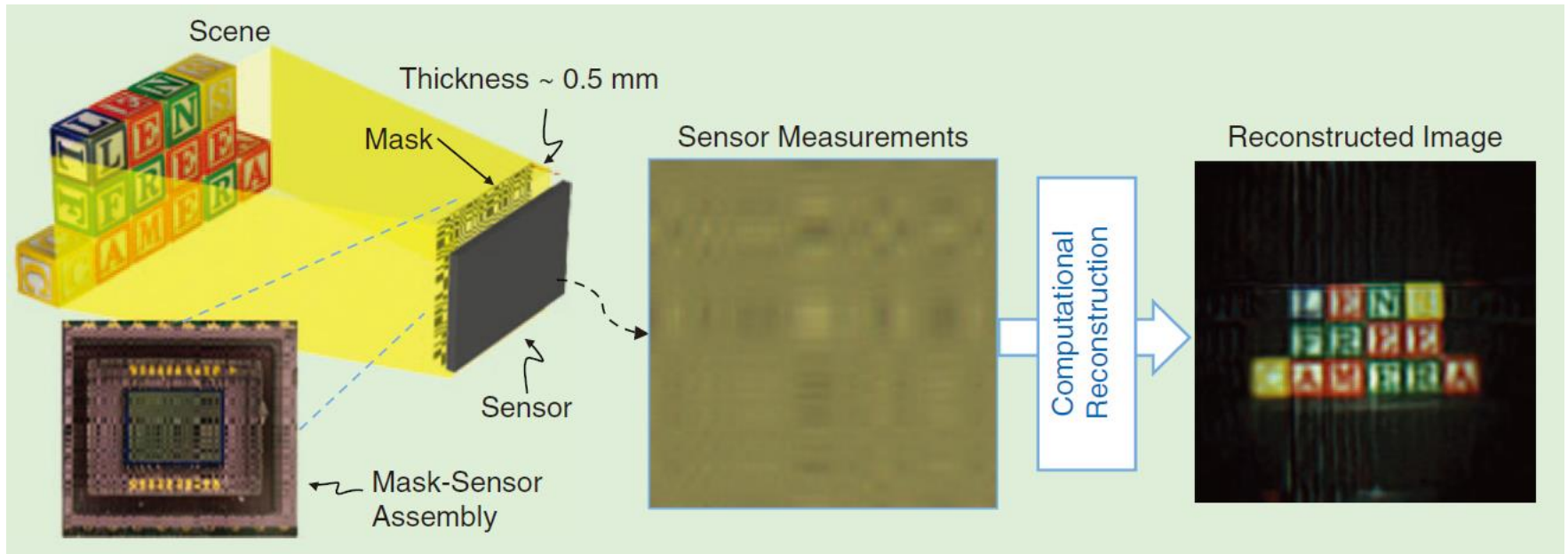
Lensless Imaging for flat/small dimension



(a)

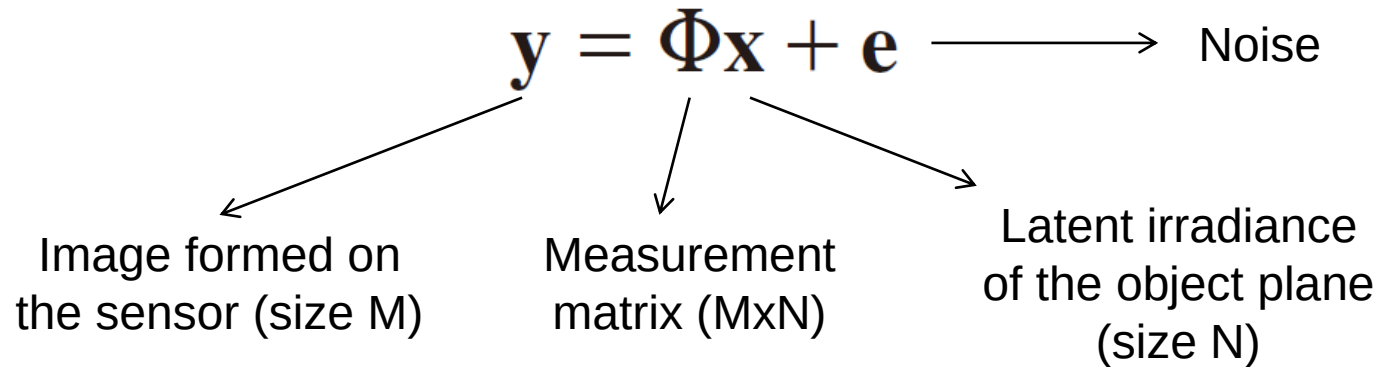


Application: FlatCam



- Image reconstruction through matrix solver
- Pseudo-random mask
- Calibration

Image formulation



- Measurement matrix is determined by the mask pattern, mask-sensor spacing, and **diffraction**.
- Use calibration to derive it, instead of theoretical analysis, due to high modeling complexity.



Image reconstruction

Regularized inverse problem

$$\min_{\mathbf{x}} \|\mathbf{y} - \Phi \mathbf{x}\|_2^2 + \lambda \mathcal{R}(\mathbf{x})$$

with total-variation (TV) regularization
for convex optimization

$$\mathcal{R}(\mathbf{x}) = \|\mathbf{x}\|_{\text{TV}} = \sum_{u,v} \sqrt{\mathbf{g}_u(u,v)^2 + \mathbf{g}_v(u,v)^2}$$

$$\mathbf{g}_u = D_u \mathbf{x}$$

$$\mathbf{g}_v = D_v \mathbf{x}$$



Mask design and calibration

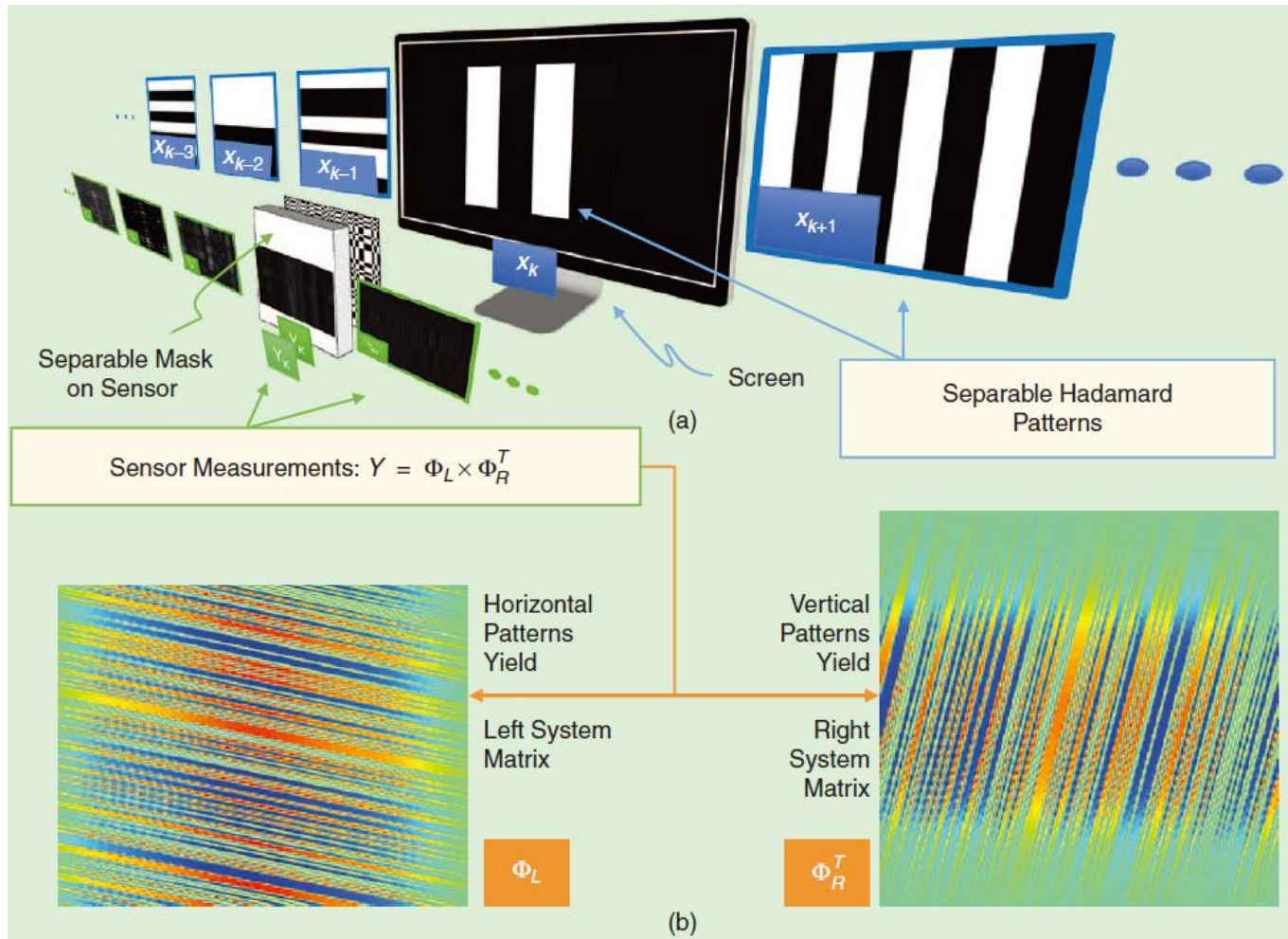
- Pseudo-random pattern
 - Keep the rank of Φ as high as possible to increase solution quality
- Separable pattern
 - Reduce the storage of Φ and also the complexity of its calibration
- Calibration
 - Measure images for known separable scene X

$$X = \mathbf{a}\mathbf{b}^T$$

$$Y = \Phi_L \mathbf{a}\mathbf{b}^T \Phi_R^T = (\Phi_L \mathbf{a})(\Phi_R \mathbf{b})^T$$

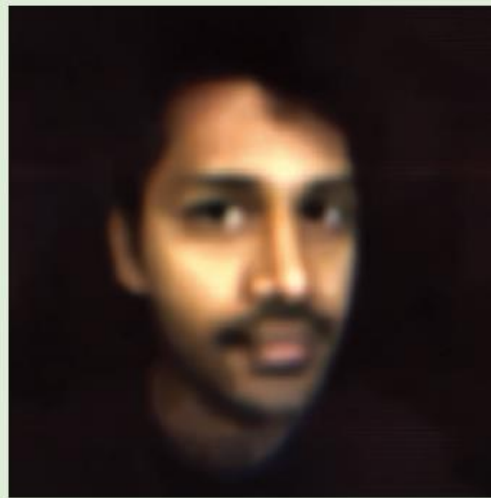
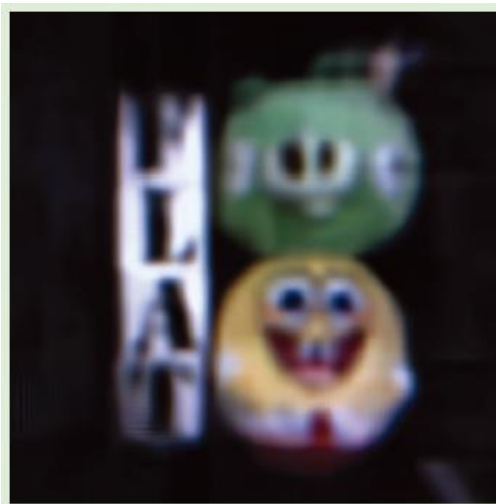
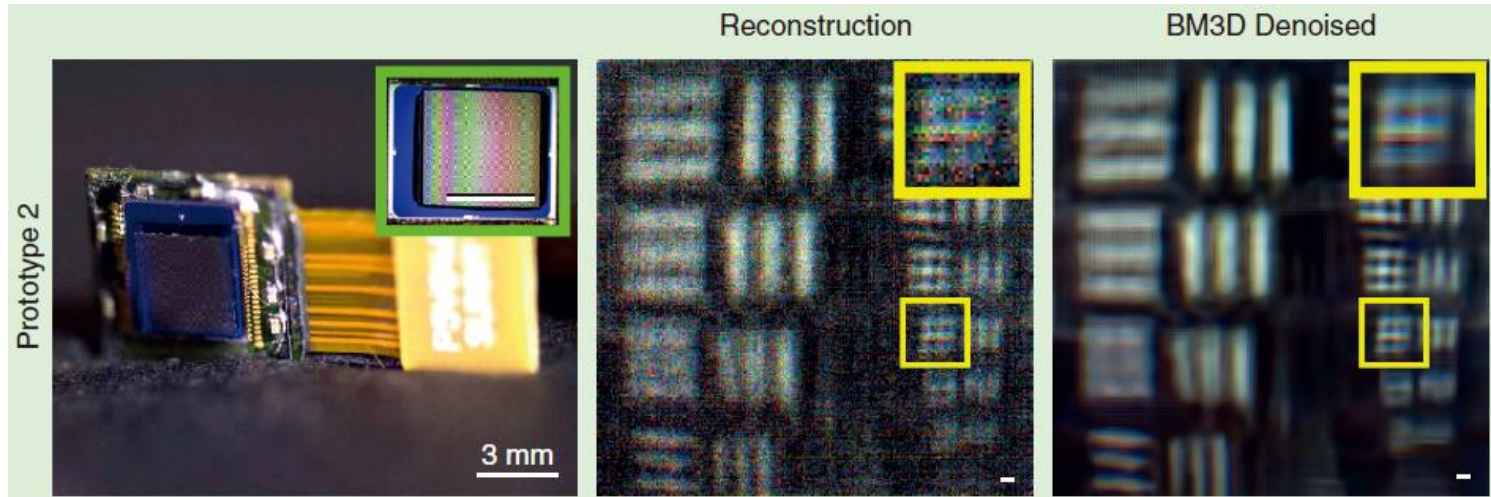
Rank-1 decomposition by SVD

Calibration

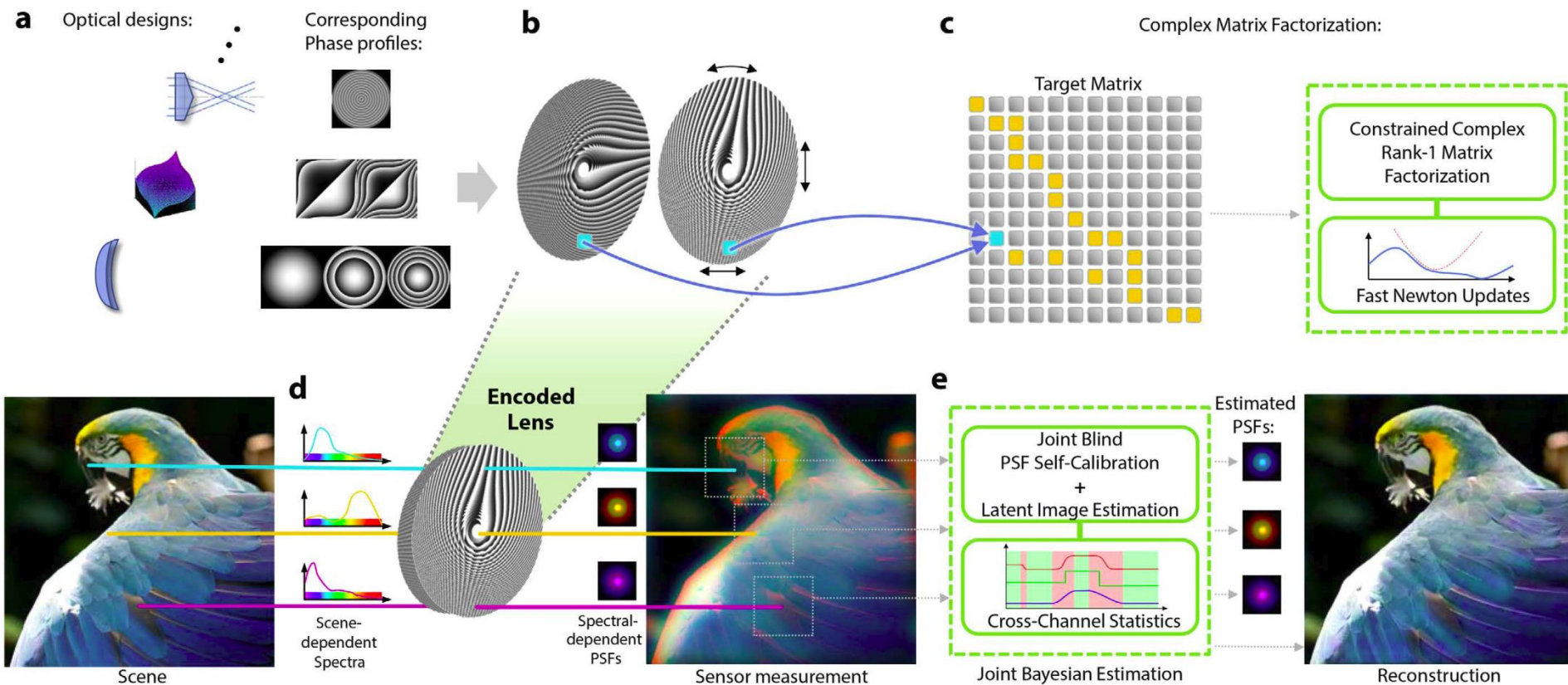


Note that one measurement matrix Φ is only for one object depth.

Result



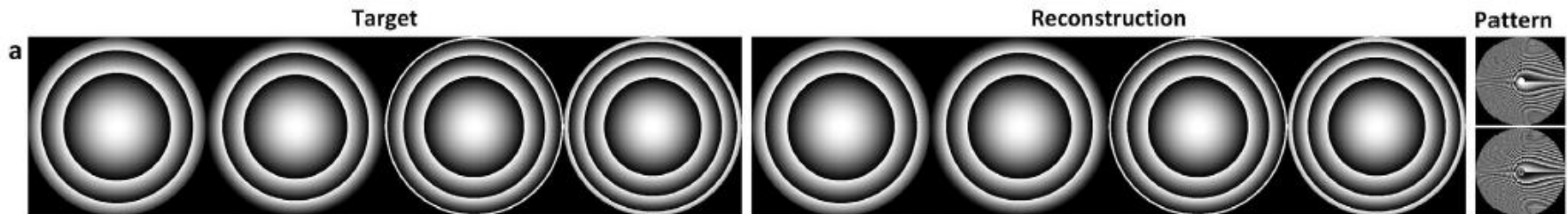
Application: Diffractive optics for true-color imaging



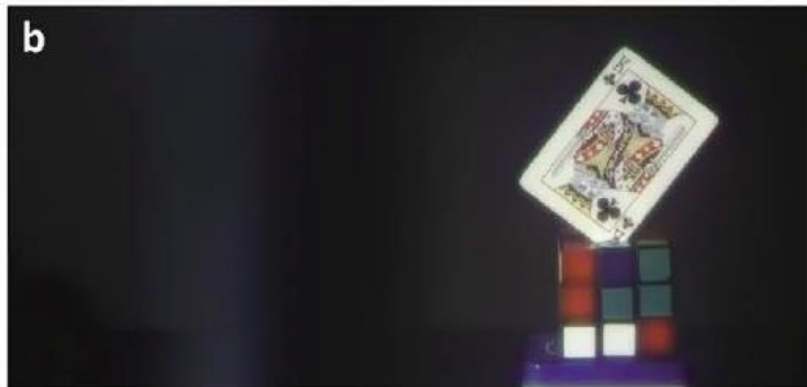
Ref: F. Heide, et. al., "Encoded diffractive optics for full-spectrum computational imaging," *Scientific Reports.*, Sept 2016.

Encoded diffractive lens

- A stack of two static diffractive optic elements (DOEs) can produce a Rank-1 approximation of a complex matrix



DOE rotation encodes focal length



Focus at far distance



Focus at near distance

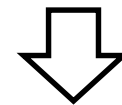


Self-calibrated blind deconvolution

- Optimization formulation

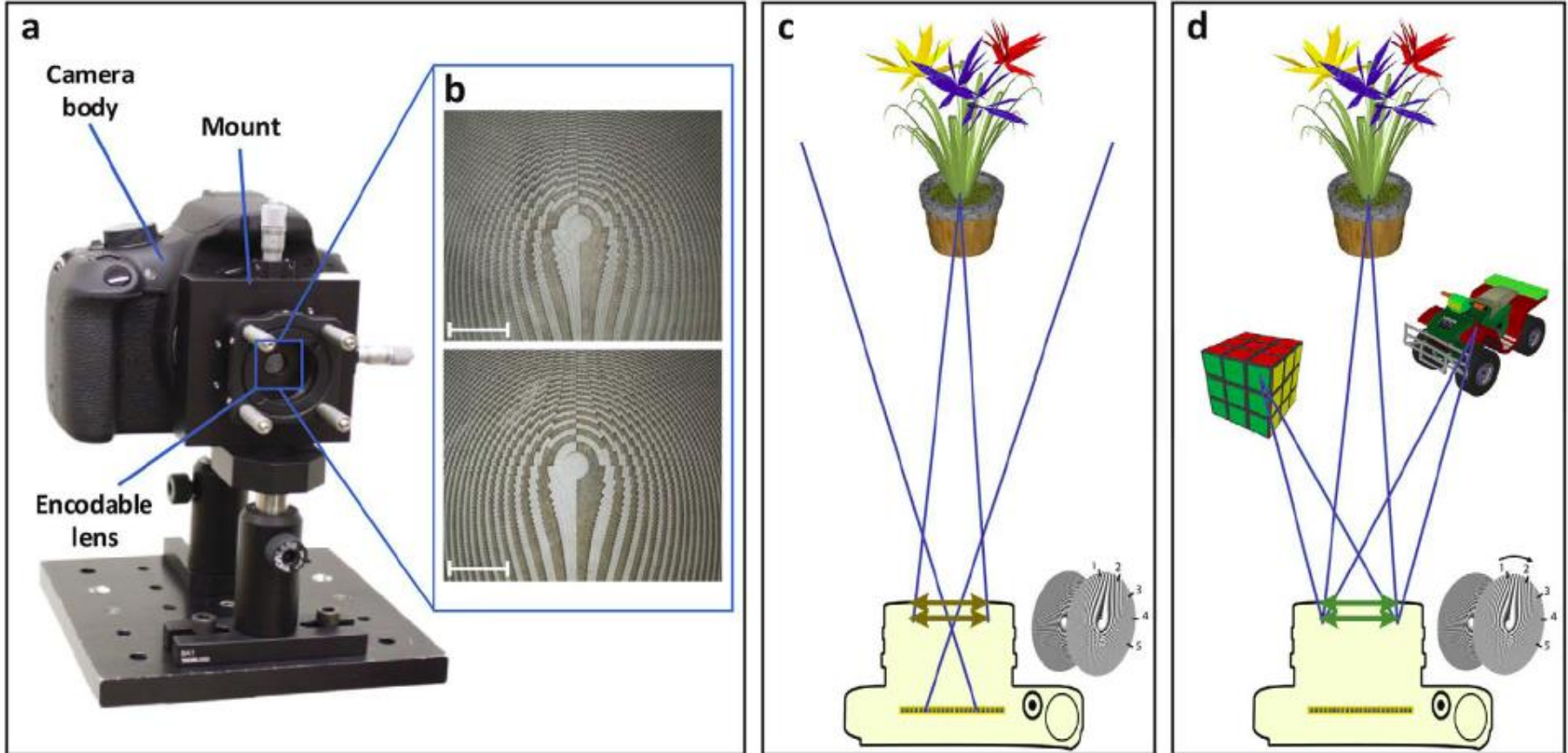
$$\begin{aligned} \mathbf{x}_{\text{opt}}, \mathbf{v}_{\text{opt}} = & \underset{\mathbf{x}, \mathbf{v}}{\operatorname{argmin}} \underbrace{\|\mathbf{x} * \mathbf{v} - \mathbf{J}\|_2^2}_{\text{data term}} + \underbrace{\alpha \sum_{a=1}^3 \|\mathbf{h}_a * \mathbf{x}\|_2^2}_{\text{L2 regularization term for PSF}} + \underbrace{\beta \|\mathbf{x}\|_2^2}_{\text{Low-energy regularization term for PSF}} \\ & + \underbrace{\gamma \sum_{a \neq r} \|\mathbf{h}_a * \mathbf{v} - \mathbf{h}_a * \mathbf{I}_r\|_1}_{\text{Cross-channel color prior}} + \underbrace{\mu \sum_{a=1}^3 \|\mathbf{h}_a * \mathbf{v}\|_1}_{\text{L1 regularization term for latent image}}, \end{aligned}$$

x: spatially-varying PSFs
v: latent image
h: gradient operator

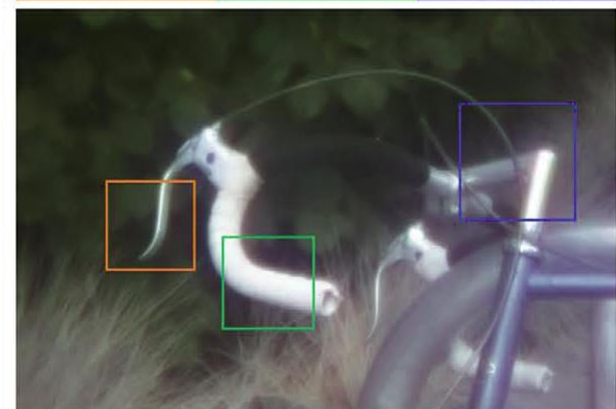
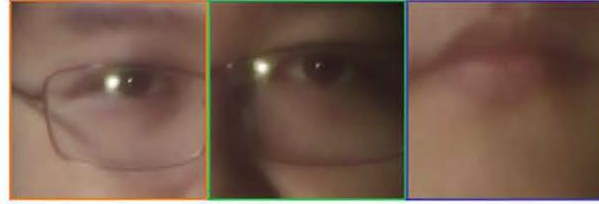
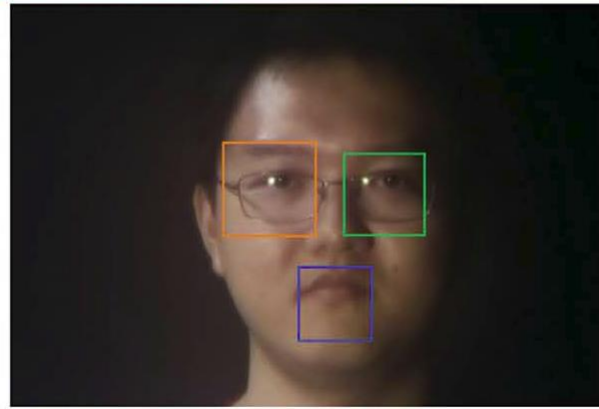
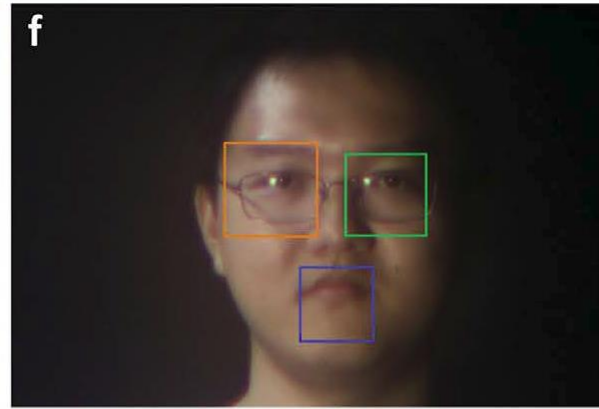
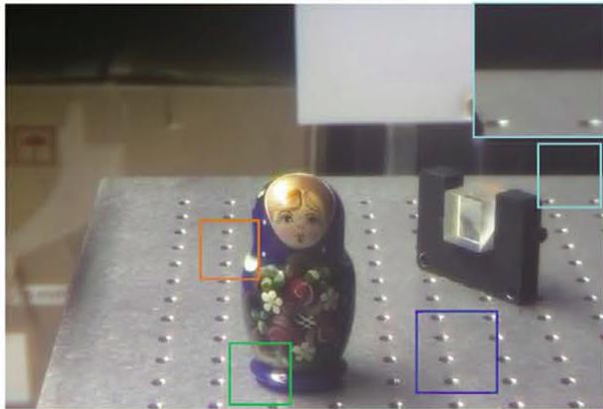
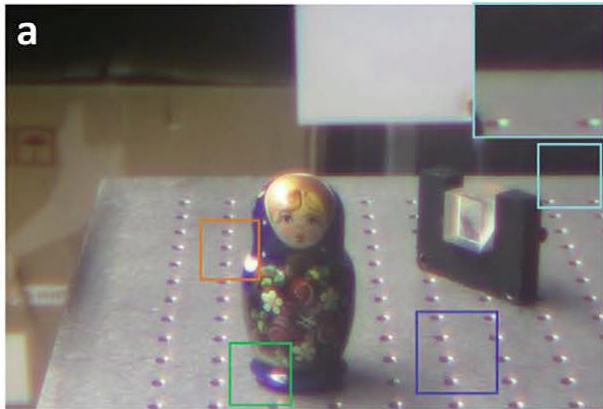


Solved by ProxImaL

Experimental setup



Captured



Reconstructed